

Gossip and Push-Sum convergence

COP6539 Project 2 · Ahmed Rageeb Ahsan, Saiful Islam

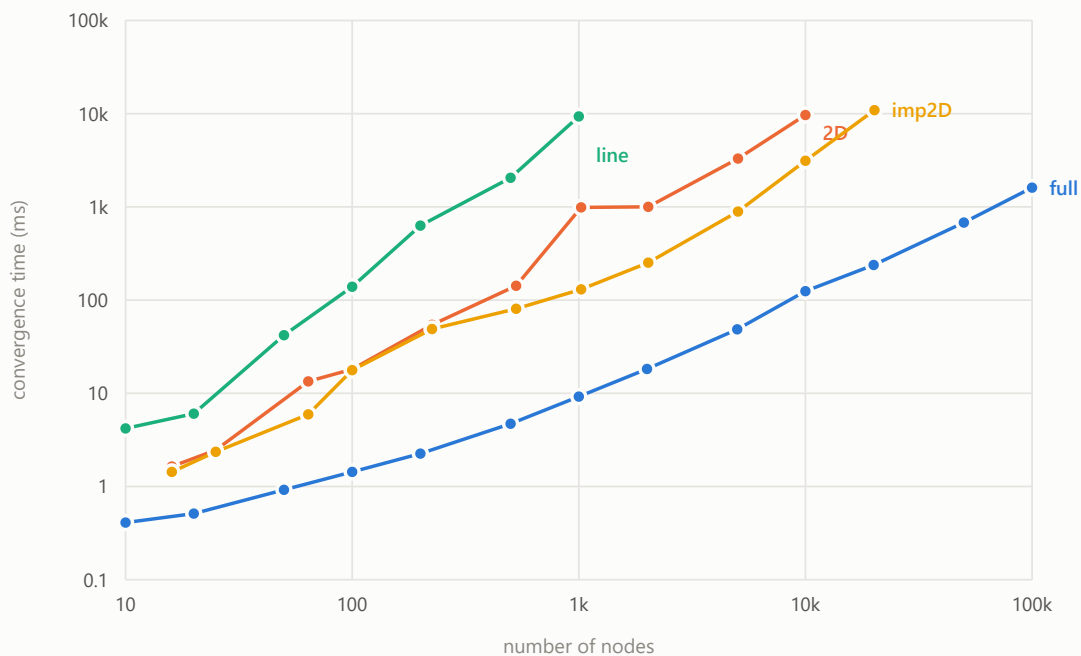
An Erlang actor simulator measuring how long Gossip and Push-Sum take to converge across four network topologies. Every node is one actor; the only coordination is asynchronous message passing. Each point is the median of five trials, counting only the trials that converged.

1. Convergence time vs network size

■ full ■ 2D ■ line ■ imp2D

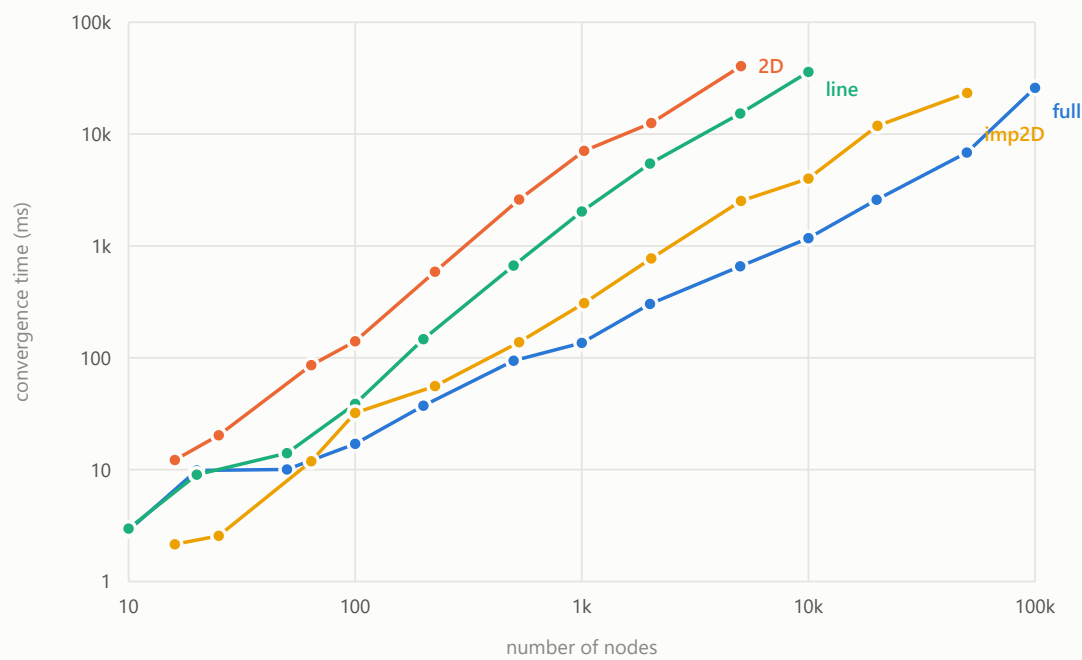
Gossip

Median over converged trials. Both axes logarithmic.



Push-Sum

Median over converged trials. Both axes logarithmic.



2. Largest network reached

Topology	Gossip — largest n	time	Push-Sum — largest n	time
full	100000	1608.5 ms	100000	25891.2 ms
2D	10000	9675.3 ms	5041	40510.9 ms
line	1000	9339.8 ms	10000	35944.2 ms
imp2D	20164	10910.7 ms	50176	23312.8 ms

3. Measurements

Algorithm	Topology	Nodes	Median (ms)	Converged
Gossip	full	10	0.41	5/5
Gossip	full	20	0.51	5/5
Gossip	full	50	0.92	5/5
Gossip	full	100	1.43	5/5
Gossip	full	200	2.25	5/5
Gossip	full	500	4.71	5/5
Gossip	full	1000	9.22	5/5
Gossip	full	2000	18.23	5/5
Gossip	full	5000	48.54	5/5
Gossip	full	10000	124.62	5/5

Algorithm	Topology	Nodes	Median (ms)	Converged
Gossip	full	20000	237.88	5/5
Gossip	full	50000	678.91	5/5
Gossip	full	100000	1608.50	5/5
Gossip	2D	16	1.64	5/5
Gossip	2D	25	2.46	5/5
Gossip	2D	64	13.41	5/5
Gossip	2D	100	18.13	5/5
Gossip	2D	225	54.27	5/5
Gossip	2D	529	142.34	5/5
Gossip	2D	1024	987.24	5/5
Gossip	2D	2025	1000.81	4/5
Gossip	2D	5041	3298.71	3/5
Gossip	2D	10000	9675.26	5/5
Gossip	line	10	4.20	5/5
Gossip	line	20	6.04	5/5
Gossip	line	50	41.98	5/5
Gossip	line	100	139.01	4/5
Gossip	line	200	629.55	3/5
Gossip	line	500	2052.40	1/5
Gossip	line	1000	9339.80	1/5
Gossip	imp2D	16	1.43	5/5
Gossip	imp2D	25	2.35	5/5
Gossip	imp2D	64	5.94	5/5
Gossip	imp2D	100	17.71	5/5
Gossip	imp2D	225	48.95	5/5
Gossip	imp2D	529	80.69	5/5
Gossip	imp2D	1024	130.36	3/5
Gossip	imp2D	2025	252.11	5/5
Gossip	imp2D	5041	891.65	4/5
Gossip	imp2D	10000	3123.05	4/5
Gossip	imp2D	20164	10910.72	1/5
Push-Sum	full	10	2.87	5/5
Push-Sum	full	20	9.83	5/5
Push-Sum	full	50	10.04	5/5
Push-Sum	full	100	17.00	5/5
Push-Sum	full	200	37.27	5/5

Algorithm	Topology	Nodes	Median (ms)	Converged
Push-Sum	full	500	94.11	5/5
Push-Sum	full	1000	135.78	5/5
Push-Sum	full	2000	302.80	5/5
Push-Sum	full	5000	657.51	5/5
Push-Sum	full	10000	1173.40	5/5
Push-Sum	full	20000	2588.26	5/5
Push-Sum	full	50000	6836.22	5/5
Push-Sum	full	100000	25891.22	5/5
Push-Sum	2D	16	12.19	5/5
Push-Sum	2D	25	20.27	5/5
Push-Sum	2D	64	85.91	5/5
Push-Sum	2D	100	140.39	5/5
Push-Sum	2D	225	588.08	5/5
Push-Sum	2D	529	2599.94	5/5
Push-Sum	2D	1024	7076.15	5/5
Push-Sum	2D	2025	12543.80	5/5
Push-Sum	2D	5041	40510.87	5/5
Push-Sum	line	10	2.97	5/5
Push-Sum	line	20	9.01	5/5
Push-Sum	line	50	14.03	5/5
Push-Sum	line	100	38.71	5/5
Push-Sum	line	200	146.64	5/5
Push-Sum	line	500	667.34	5/5
Push-Sum	line	1000	2033.05	5/5
Push-Sum	line	2000	5448.40	5/5
Push-Sum	line	5000	15276.54	4/5
Push-Sum	line	10000	35944.19	2/5
Push-Sum	imp2D	16	2.15	5/5
Push-Sum	imp2D	25	2.56	5/5
Push-Sum	imp2D	64	11.88	5/5
Push-Sum	imp2D	100	32.15	5/5
Push-Sum	imp2D	225	55.81	5/5
Push-Sum	imp2D	529	137.83	5/5
Push-Sum	imp2D	1024	308.22	5/5
Push-Sum	imp2D	2025	773.12	5/5
Push-Sum	imp2D	5041	2525.70	5/5

Algorithm	Topology	Nodes	Median (ms)	Converged
Push-Sum	imp2D	10000	3999.85	5/5
Push-Sum	imp2D	20164	11855.57	5/5
Push-Sum	imp2D	50176	23312.79	5/5

5. Findings

Connectivity dominates everything else. At every network size the four topologies separate by orders of magnitude, in the order *full* < *imp2D* < *2D* < *line*. Adding a single random long-range edge per node — the only difference between 2D and imp2D — is worth far more than the regular grid structure it is added to. That is the small-world effect appearing in a 25-line change to the neighbour function.

Gossip on a line is a coin flip, not a slow success. The rumour does not simply take longer on a line; it frequently dies out altogether. A node stops transmitting after ten hears, and the node adjacent to a fast talker reaches ten almost immediately, having forwarded the rumour only a handful of times. The frontier then stalls, and what remains is a dead-end node transmitting forever into a neighbour that has already fallen silent. This is why the tables report how many trials converged rather than a bare average.

Push-Sum is accurate exactly where it is fast. On full and imp2D the estimate matches the true average to around $1e-13$ — floating-point noise. On 2D and line the relative error rises to roughly $1e-3$. The termination rule is *local*: an actor stops when its own ratio stops moving, which on a slowly-mixing graph happens well before the network as a whole has mixed. Accuracy and speed are not two independent properties here; they are the same property of the topology.

On what *s/w* converges to. With $s_i = i$ and $w_i = 1$, the conserved quantity is $\sum s / \sum w = (n+1)/2$ — the *average*, not the sum. The assignment calls *s/w* the sum estimate; the sum is the ratio times n . Both are printed by the simulator.

Scaling. Across 10 to 100000 nodes, gossip on the full topology fits $t \propto n^{0.90}$. A slope near 1 means convergence time grows roughly linearly with the number of actors, which is what a fixed number of rounds costs when each round is $O(n)$ messages spread over a fixed number of cores — the simulator is scheduler-bound, not round-bound.